

1) for the following sequence of pages using FIFO, LRU, MRU & optimal page replacement algorithms.

Sequence: 7 0 1 2 0 3 0 4 2 3 0 3 0 8 2 1

NO. OF frames in physical memory are: 4
Given,

→ Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 0 3 2 1

→ NO. OF frames: 4

li) FIFO - First in First Out

In FIFO, we replace the page that entered the memory first.

step	page	F ₁	F ₂	F ₃	F ₄	Result
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	2	7	0	2	-	F
4	0	7	0	2	-	H
5	3	7	0	2	3	F
6	0	7	0	2	3	H
7	4	4	0	2	3	F
8	2	4	0	2	3	H
9	3	4	0	2	3	H
10	0	4	0	2	3	H
11	3	4	0	2	3	H
12	2	4	0	2	3	H
13	1	1	0	2	3	F

page faults = 6

page Hits = 7

(ii) LRU - Least Recently used.

In LRU, we replace the page that has not been used for the longest time.

Step	page	F1	F2	F3	F4	Result
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	2	7	0	2	-	F
4	0	7	0	2	3	H
5	3	7	0	2	3	F
6	0	7	0	2	3	H
7	4	4	0	2	3	F
8	2	4	0	2	3	H
9	3	4	0	2	3	H
10	0	4	0	2	3	H
11	3	4	0	2	3	H
12	2	4	0	2	3	H
13	1	4	0	2	1	F

Page Faults = 6

Page Hits = 7

(iii) MRU - Most Recently Used

In MRU, we replace the page that was used most recently.

step	page	F1	F2	F3	F4	Result
1	7	7	-	-	-	F
2	0	7	0	0	-	F
3	2	7	0	2	-	F
4	0	7	0	2	3	F
5	3	7	0	2	3	H
6	0	7	0	2	3	F
7	4	7	4	2	3	H
8	2	7	4	2	3	H
9	3	7	4	2	3	H
10	0	7	4	2	3	F
11	3	7	4	2	3	F
12	2	7	4	2	3	H
13	1	7	4	1	3	H

page faults = 8

page hits = 5

(iv) Optimal page replacement.

In optimal page, replace the page whose next use is farthest in the future

Step	Page	F1	F2	F3	F4	Result
1	7	7	-	-	-	F
2	0	7	0	-	-	F
3	2	7	0	2	-	H
4	0	7	0	2	-	F
5	3	7	0	2	3	H
6	0	7	0	2	3	F
7	4	7	0	2	4	H
8	2	7	0	2	4	H
9	3	7	0	2	4	H
10	0	7	0	2	4	H
11	3	7	0	2	4	H
12	2	7	0	2	4	F
13	1	7	0	2	4	F

Page faults = 6

Page hits = 7

Therefore, total reference string = 13 pages

So;

FIFO = 6 faults

LRU = 6 faults

MRU = 8 faults

Optimal = 6 faults

2) Apply FIFO page replacement algorithm for the following sequence of pages and check whether the belady's / anomaly exists or not. For checking up belady anomaly take the physical memory size as 3 frames and 4 frames.

Page sequence: 1 2 3 4 1 2 5 1 2 3 4 5

Implementation of 3 algorithms i.e., checking of efficiency, Belady's algorithm & anomaly efficiency.

Given,

Reference string: 1 2 3 4 1 2 5 1 2 3 4 5.

we need to check with 3 frames and 4 frames.

With 3 Frames

1 2 3 → fault

4 replaces 1 → fault

1 replaces 2 → fault

2 replaces 3 → fault

5 replaces 4 → fault

1 → Hit

2 → Hit

3 replaces 1 → fault

4 replaces 2 → fault

5 → Hit

Total Page Faults = 9

* With 4 frames

1 2 3 4 $\rightarrow$ 4 faults

1 $\rightarrow$ Hit

2 $\rightarrow$ Hit

5 replaces 1 $\rightarrow$ Fault

1 replaces 2 $\rightarrow$ Fault

2 replaces 3 $\rightarrow$ Fault

3 replaces 4 $\rightarrow$ Fault

4 replaces 5 $\rightarrow$ Fault

5 replaces 1 $\rightarrow$ Fault

Total page faults = 10.

Frames	FIFO page faults
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3 Frames	9
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4 Frames	10
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since page faults increase from

9 $\rightarrow$ 10 when frames increase from
3 $\rightarrow$ 4, Belady's Anomaly also exists.